Lipoxygenases from diatoms

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Poster

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AIMS: Due to the role in environment, we investigated lipoxygenase (LOX) structure in diatoms by integrating bioinformatics tools

1.DATABASE
GenBank
NCBI Protein
Uniprot
TSA

→

2.ALIGNMENT
Blast
Clustal Omega
T-coffee

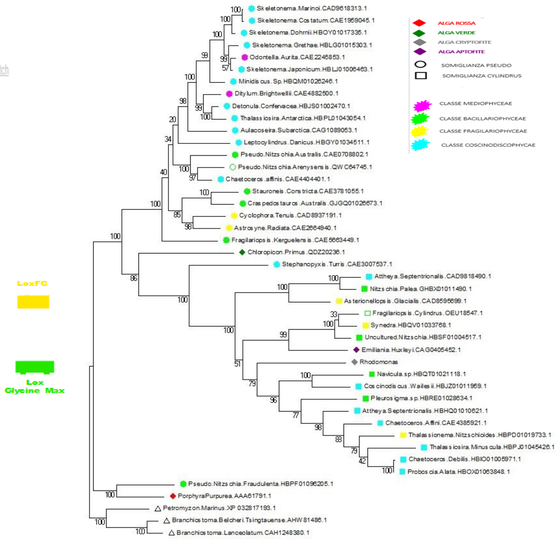
→

3.PHILOGENETIC TREE
Mega 6

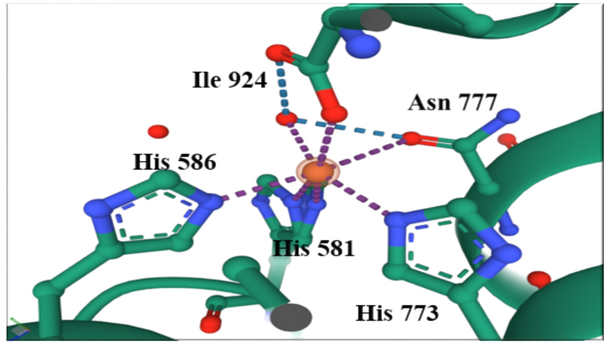
→

4. STRUCTURE MODELING
Swiss-Model
Modeller
Pymol

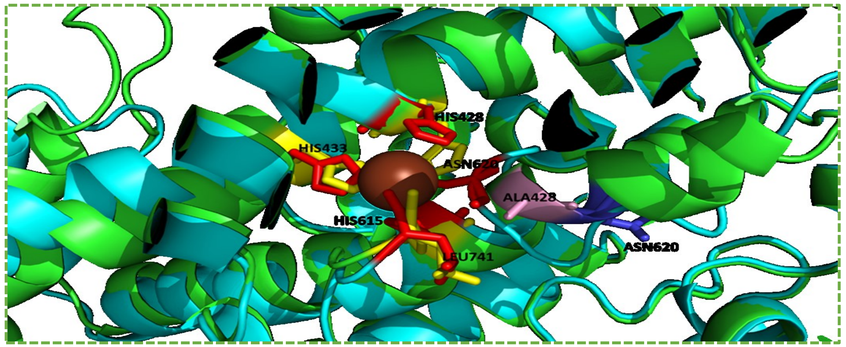
Sequence comparison and phylogenetic analyses evidenced pecularities and possible evolutive differences



The conservation among diatoms of amino acids found in the catalytic site and responsible of the coordination of the Fe^{2+} cofactor, guided the 3D modelling and comparison with available RX and AlphaFold models. Their correct position in coordinating the cofactor confirms the quality of our models.



Structure of catalytic site of LOX from Oryza sativa



Comparion of structures from AlphaFold and from our modelling procedure